

Michael Sagar Vasandani — Résumé

San Diego, California

Email Michael Michael Vasandani on GitHub LinkedIn profile

Experience

AI Engineer Intern — ResMed

San Diego, CA · June 2026 – Present

- Cut bug-fix turnaround by 60% by engineering a production multi-agent Crashlytics remediation workflow with GitHub Agentic Workflows for automated investigation, fix generation, and human-in-the-loop validation.
- Enabled 40+ users to self-serve internal questions by building a LangChain/LangGraph AI investigation platform with MCP tools, RAG memory, a FastAPI REST API, and AWS CloudWatch/DynamoDB integrations.
- Increased internal AI adoption by 70% by launching a reusable agent marketplace with CI/CD validation and versioning.
- Saved 10+ developer hours weekly with an agentic workflow that detects documentation drift and opens pull requests.

Software Engineer - AI Intern — Qualcomm Institute

San Diego, CA · September 2025 – May 2026

- Cut content authoring time by 85% with an LLM generation/compliance pipeline using FastAPI, MongoDB, and Ollama.
- Integrated the system into Unreal Engine via C++ components, enabling real-time API calls and bulk dialogue importing.
- Reduced token usage and processing time by 70% by implementing semantic retrieval over prior dialogue trees in MongoDB.

Research Assistant — Smarr Lab

San Diego, CA · April 2025 – May 2026

- Built an AI knowledge base using Python, Codex, and Obsidian that transforms academic papers into knowledge graphs.
- Accelerated an ETL pipeline 7x for 700M+ rows by optimizing BigQuery SQL and leveraging Polars for parallel processing.

Software Engineer Intern — Boost Health Labs

San Diego, CA · June 2025 – September 2025

- Increased website engagement by 45% by building Angular/TypeScript components and integrating REST services.
- Owned full-stack features from requirements through production by collaborating across product and development teams.

Software Engineer Intern — TipTop Technologies

Sunnyvale, CA · March 2025 – June 2025

- Architected internal GitHub analytics tool on AWS Lambda and DynamoDB, cutting leadership review time by 90%.
- Engineered LLM pipeline with RAG and Pinecone embeddings for natural language codebase queries.

Publication Systems Engineer — Fixture Laboratory

Remote · January 2024 – February 2025

- Designed a deterministic publication verifier that compares every rendered field against immutable evidence before promotion.
- Documented recovery behavior across retries, provider timeouts, and last-valid deployment restoration without hiding operational failures.

Education

University of California, San Diego - San Diego, CA

Bachelor of Science in Data Science - GPA: 3.82

June 2026

Coursework: Algorithms, Object-Oriented Programming, ML/AI, Data Structures, Databases, Data Visualization

Projects

Hackathon-In-A-Box

TypeScript, Claude Agent SDK, Node.js, PostgreSQL

[View Hackathon-In-A-Box repository](#)

- Placed 2nd at the Claude Social Impact Hackathon by building an 8-phase agentic planning platform that generates source-verified venue, sponsor, and mentor recommendations.
- Drove adoption across 10+ nonprofits by conducting stakeholder and user interviews, translating feedback into product requirements, and iterating on the platform to match real-world planning workflows.

SafeTrip SF

Python, LangGraph, Pydantic, FastAPI, PostgreSQL, pgvector

[View SafeTrip SF repository](#)

- Won the Musa Labs Hackathon by building a RAG-powered trip planner generating itineraries from safety data.
- Built with LangGraph, FastAPI, React, PostgreSQL, and pgvector using 10,000+ incidents across 41 neighborhoods.

Personal Call Agent

TypeScript, Vercel, Google Calendar API, Bland AI

[View Personal Call Agent repository](#)

- Built a real-time AI voice agent that handles missed calls, checks availability, books meetings, and sends call summaries.
- Integrated Bland AI with Google Calendar for scheduling, with webhook verification and API contract testing via Vitest.

Technical Skills

Languages: Java, Python, C++, SQL, JavaScript, TypeScript, HTML/CSS, R

Frameworks: React, Node.js, Angular, Material-UI, FastAPI

Tools & Cloud: Git, GitHub Actions, Docker, Kubernetes, AWS, GCP, LangGraph, Ollama, Pinecone, pgvector

Libraries: pandas, NumPy, scikit-learn, PyTorch, TensorFlow, LangChain, Hugging Face Transformers, Pydantic

Awards

- Second Place, Claude Social Impact Hackathon
- Winner, Musa Labs Hackathon

Certifications

- AWS Certified Cloud Practitioner

Publications

- Reliable evidence graphs for agentic publication systems — engineering note, 2026

Volunteering

- Mentor, student AI engineering workshop series

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